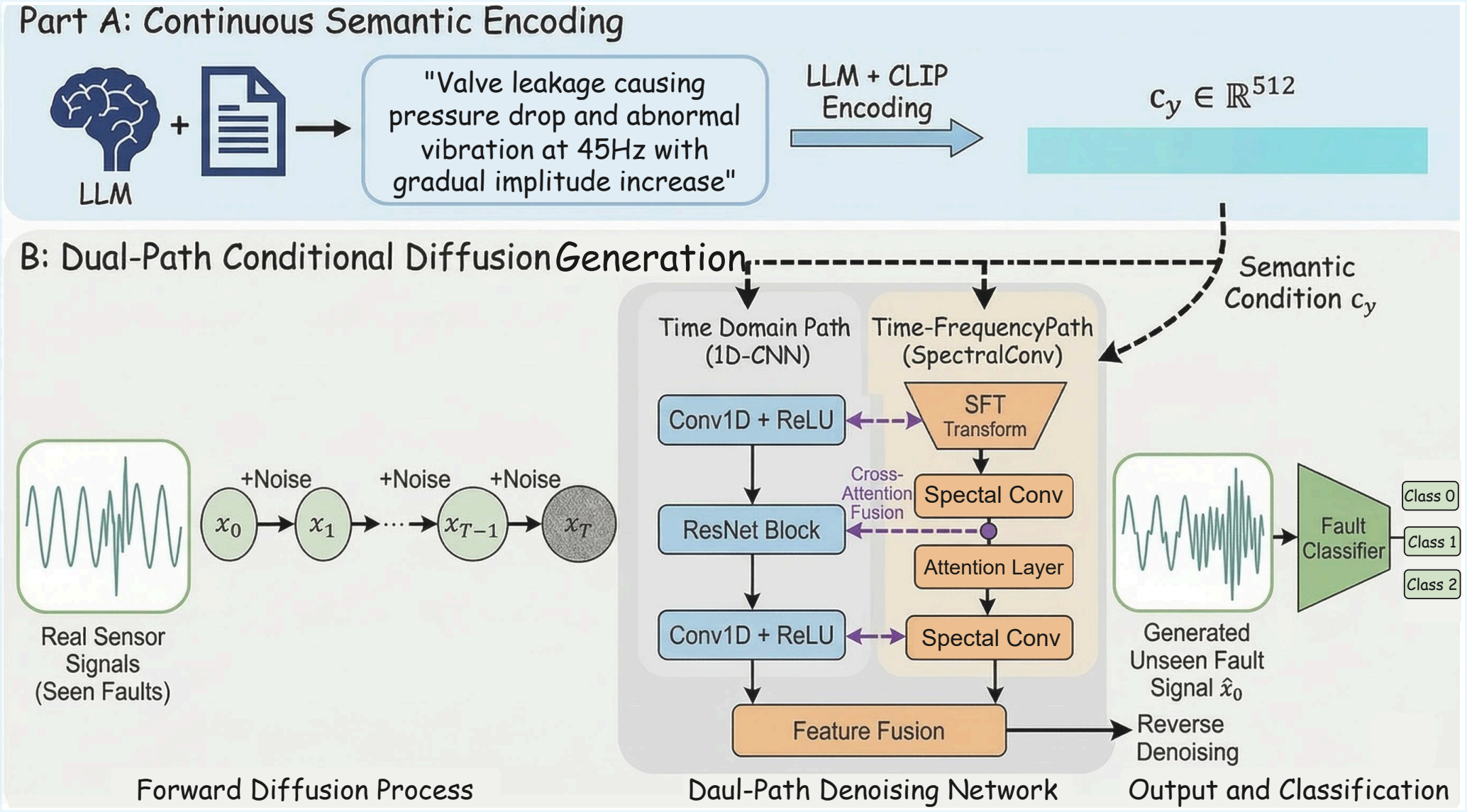
Part A: Continuous Semantic Encoding
LLM
"Valve leakage causing pressure drop and abnormal vibration at 45Hz with gradual implitude increase"
LLM + CLIP Encoding
$c_y \in \mathbb{R}^{512}$
B: Dual-Path Conditional Diffusion Generation
Semantic Condition c_y
Time Domain Path (1D-CNN)
Time-FrequencyPath (SpectralConv)
Conv1D + ReLU
SFT Transform
Cross-Attention Fusion
Spectal Conv
ResNet Block
Attention Layer
Conv1D + ReLU
Spectal Conv
Feature Fusion
Reverse Denoising
Real Sensor Signals (Seen Faults)
+Noise
+Noise
+Noise
x_0
x_1
x_{T-1}
x_T
Generated Unseen Fault Signal $\hat{x}_0$
Fault Classifier
Class 0
Class 1
Class 2
Forward Diffusion Process
Daul-Path Denoising Network
Output and Classification